Zhuodong Li

Education

2023 - 2026 **M.S. in Computer Graphics** University of Chinese Academy of Sciences GPA: 3.63/4

2019 - 2023 **B.S. in Digital Media Technology** Shandong University GPA: 3.61/4

Publication

Li, **Zhuodong**, Fei Hou, Wencheng Wang, Xuequan Lu, and Ying He (July 2025). "A Divide-and-Conquer Approach for Global Orientation of Non-Watertight Scene-Level Point Clouds Using 0-1 Integer Optimization". In: 44.4. ISSN: 0730-0301. DOI: 10.1145/3730923. URL: https://doi.org/10.1145/3730923.

Li, **Zhuodong**, Zengke Liu, Fei Hou, Wencheng Wang, and Ying He (2026). "MegaNorm: Local Patch Embeddings for Efficient and Robust Point Normal Orientation at Super-Large Scale". In: *ACM Transactions on Graphics*. SIGGRAPH 2026, to appear.

Research Experience

Scene-level Point Cloud Orientation July 2023 – January 2025

Proposed a divide-and-conquer framework for orienting large-scale scene point clouds

- Partitioned the input cloud into patches and refined normals within each patch using propagation and iterative Poisson reconstruction
- Formulated a Boolean programming model based on the principle of visible surface consistency to achieve global orientation consistency
- Achieved significant accuracy improvements over state-of-the-art methods under varying sparsity and noise levels on ScanNet and SceneNN

Super large scale Point Cloud Orientation September 2025 – Current

- Utilized PointTransformer to predict per-point orientation for large-scale 3D data
- Achieved performance significantly surpassing current State-of-the-Art (SOTA) methods on the SceneNN test split.
- Demonstrated robust generalization through zero-shot testing on SceneNet and Tanks and Temples datasets, outperforming existing unsupervised SOTA approaches.

Awards & Honors

2025 **National Scholarship, University of Chinese Academy of Sciences**

2024 **Merit Student**, University of Chinese Academy of Sciences

2023 **First-Class Academic Scholarship**, University of Chinese Academy of Sciences

2021 **First Prize**, National College Student Mathematical Contest in Modeling (Shandong Province)

2021 **Special Talent Scholarship**, Shandong University

2020 **Special Talent Scholarship**, Shandong University

Strengths

- Strong analytical thinking and vivid imagination.
- Passionate about programming with sustained practice; proficient in C++ and Python.
- No immediate graduation pressure; available to join research groups early.
- I have a wide range of hobbies—hiking, biking, Latin dancing… There are so many things that bring me joy!